

Project Design (Updated for CSE499B Final Submission):

AI Medical Pre-Screening Assistant

From a 5-Stage ASR Pipeline to a 15-Module Conversational Clinical Intelligence System

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1 Introduction

This chapter presents the **updated** project design of the *AI Medical Pre-Screening Assistant*. The original CSE499A design was a five-stage, one-shot pipeline: Patient Speech → ASR → Medical NER → Structured EHR → Doctor Dashboard. Empirical findings from Phases 2 and 3 revealed that this linear approach was insufficient: (i) patients frequently omit critical clinical information in a single recording, (ii) there was no mechanism to handle life-threatening emergencies, and (iii) the system provided no explanation of its outputs. The design has therefore been expanded to **15 functional modules** that add an adaptive conversational follow-up loop, emergency detection, risk classification, explainable AI, and a doctor feedback and retraining pipeline. The system remains strictly a *pre-screening and decision-support tool*; all final clinical authority rests with the attending physician.

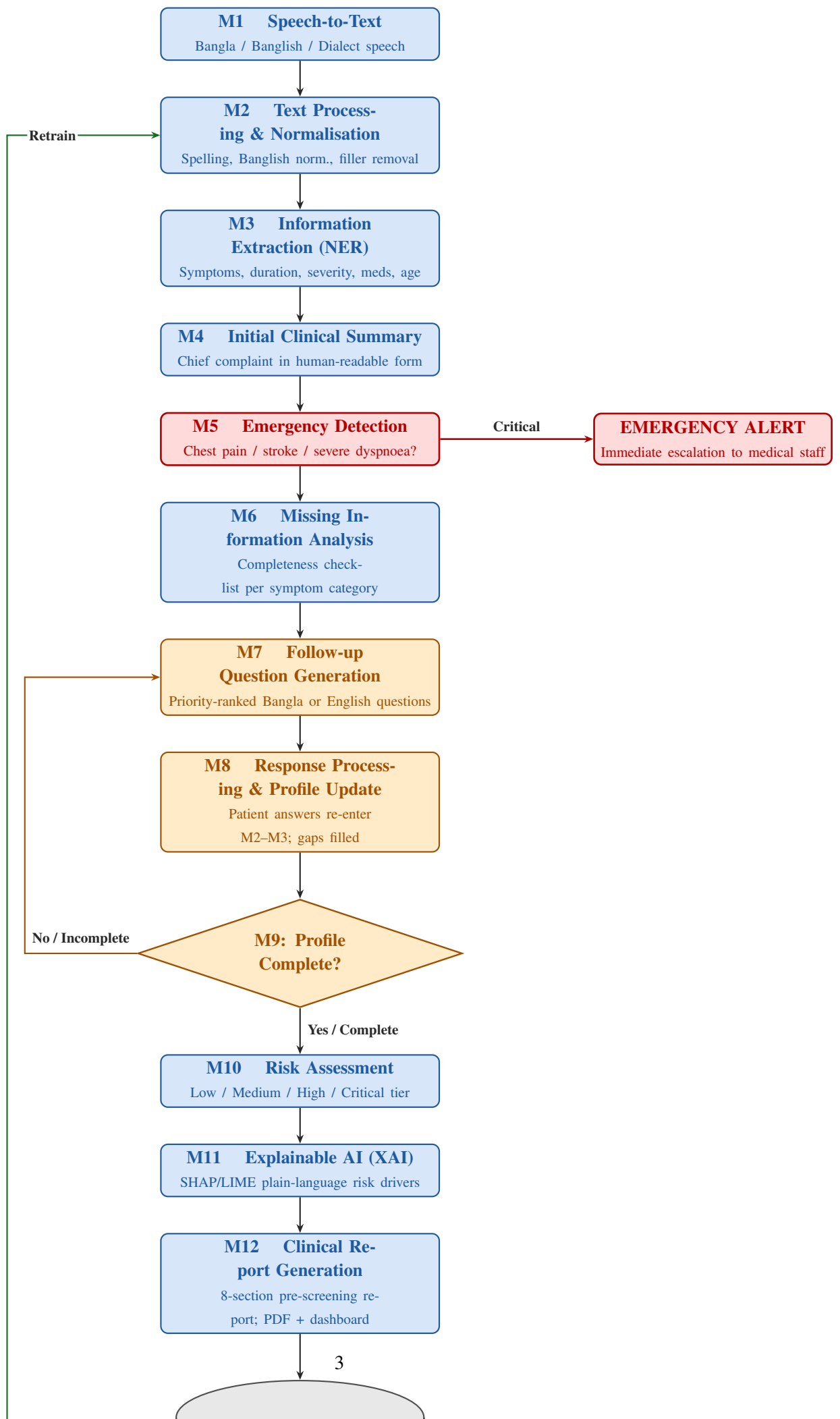
2 Design Comparison: Original vs. Updated

Dimension	Original (5 stages, CSE499A)	Updated (15 modules, CSE499B)
Interaction model	One-shot recording; no follow-up	Conversational loop; adaptive questions until profile is complete
Emergency detection	Absent	Module 5: rule-based + AI classifier; immediate escalation alert
Information gap analysis	Absent	Module 6: structured checklist per symptom category
Risk classification	Differential hints only	Module 10: Low / Medium / High / Critical with clinical rule engine
Explainability	Absent	Module 11 (XAI): SHAP/LIME plain-language driver report
Clinical report	Basic structured EHR	Module 12: full 8-section pre-screening report; PDF + dashboard
Doctor feedback loop	Absent	Module 15: annotations drive model retraining pipeline
Bangla/Banglish support	ASR & transcript only	End-to-end bilingual; follow-up questions in Bangla or English

3 Updated System Architecture — 15 Modules

3.1 Complete Pipeline Diagram

The pipeline flows **top-to-bottom** in a single main column. The adaptive follow-up loop (Modules 7–9, orange) branches to the right and returns. The emergency alert (Module 5, red) exits to the right. The feedback path (Module 15, green) returns left to Module 2.



3.2 Module Descriptions

M1 — Speech-to-Text.

Accepts Bangla, Banglish, and regional dialect speech (Dhakaiya, Barishali, Sylheti). VAD-segments audio into 10–15 s clips. Best benchmark: *BengaliAI Regional* (46.94% WER); target after CSE499B LoRA fine-tuning: <20% WER.

M2 — Text Processing & Normalisation.

Spelling correction, Banglish normalisation, filler-word removal, punctuation restoration, sentence-boundary detection. Bilingual Bangla–English throughout.

M3 — Information Extraction (NER).

BanglaBERT token classifier labels: *symptom, body site, duration, severity, frequency, age, comorbidity, medication*. Temporal expressions normalised to standard intervals.

M4 — Initial Clinical Summary.

Concise human-readable chief-complaint summary used downstream and displayed on the dashboard. Example: “*Fever + dry cough for 3 days; mild fatigue; no chest pain.*”

M5 — Emergency Detection (new).

Rule-based critical-symptom checker + AI classifier. Red flags: chest pain, stroke signs, severe dyspnoea, loss of consciousness, acute trauma. Escalates immediately; pipeline continues in parallel for non-critical cases.

M6 — Missing Information Analysis (new).

Compares extracted entities against a per-symptom clinical checklist (☑ Fever detected; ☒ Duration missing; ☒ Temperature not provided) and outputs a structured gap report.

M7 — Follow-up Question Generation (new).

Generates priority-ranked questions in Bangla or English. Example: “*Apnar jwor ki matro kal theke?*” / “*When did the fever start?*” Already-answered items are never repeated.

M8 — Response Processing & Profile Update (new).

Patient answers re-enter M2–M3. Extracted entities merge into the profile with conflict resolution; filled gaps are marked.

M9 — Case Completion Check (new).

Scores profile completeness against a configurable threshold. If incomplete → loops to M7; if complete (or max turns reached) → proceeds to M10.

M10 — Risk Assessment (new).

Assigns **Low / Medium / High / Critical** using a clinical rule engine weighted by age, comorbidities, severity, and duration. Integrates M5 output.

M11 — Explainable AI (XAI) (new).

SHAP/LIME attribution converted to plain language. Example: “*High risk: fever >5 days + dyspnoea + age >65.*” Displayed alongside every risk classification.

M12 — Clinical Report Generation (new).

8-section pre-screening report: Patient Profile, Chief Complaint, Symptoms, Emergency Flags, Q&A Log, Risk + XAI, Condition Categories, Next Steps. Exported as PDF and to the dashboard.

M13 — EHR Database.

HIPAA/PDPA-compliant PostgreSQL with encryption at rest and in transit. Full audit logging; efficient retrieval by patient ID or date.

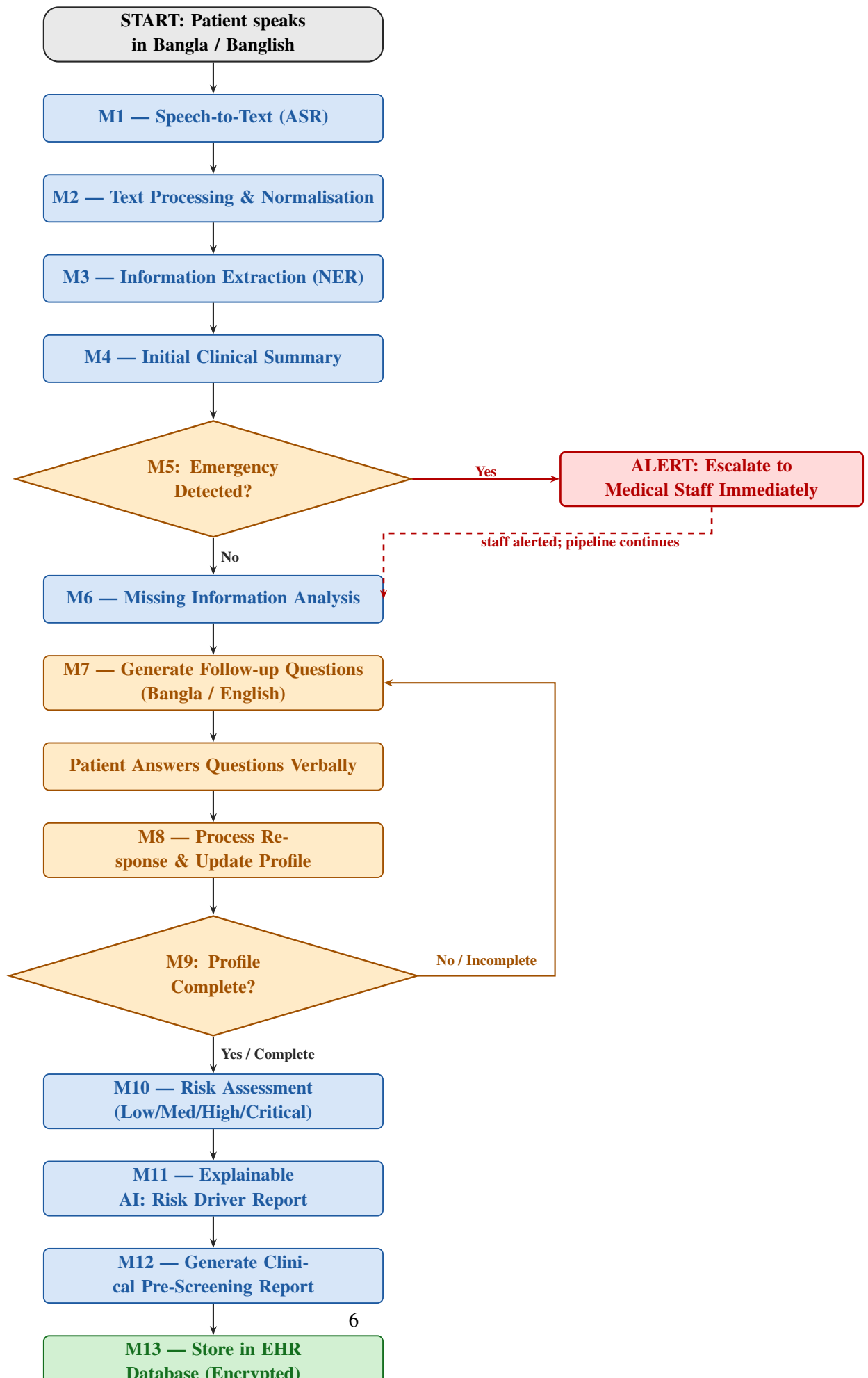
M14 — Doctor Dashboard (extended from original).

Displays report, risk tier, emergency flags, and XAI explanation. Annotation and override of any field (all changes versioned); real-time alerts for High/Critical cases.

M15 — Feedback & Continuous Learning (new).

Doctors rate reports, correct extractions, flag missed items, annotate risk disagreements. Periodic retraining of M3, M7, M10 with regression testing before deployment.

4 System Flowchart — Patient Journey



5 Software Components

Tool	Function	Rationale
Python 3.10	Implementation language	Standard ML/NLP ecosystem
PyTorch 2.x	Deep-learning framework	Native support across all benchmarked ASR models
HF Transformers	Model loading & inference	Unified API: Whisper, Wav2Vec2, MMS, Qwen-Audio, Phi-4
HF PEFT/Accelerate	LoRA fine-tuning (CSE499B)	Parameter-efficient fine-tuning of Qwen3-ASR-1.7B
BanglaBERT	Medical NER — Module 3	Pre-trained on large Bangla corpus; best Bangla NLP accuracy
WhisperX	VAD segmentation — Module 1	Handles audio >30 s with forced alignment
LIME / SHAP	XAI attribution — Module 11	Standard explainability libraries
FastAPI	Backend REST/WebSocket	Async; low latency for real-time follow-up dialogue loop
PostgreSQL	EHR storage — Module 13	Encryption-at-rest; HIPAA-ready
React / Next.js	Doctor dashboard — Module 14	Low-bandwidth; role-based access control
Flutter / PWA	Patient kiosk — Module 1	Single record button; Bangla voice prompts
MLflow	Feedback & retraining — M15	Experiment tracking; model versioning
jiwer	WER / CER / BLEU evaluation	Standard ASR metric library
Google Colab (T4/A100)	GPU compute	Free tier for inference benchmarks

6 CSE499A Implementation Phases

Phase 1 — Audio Collection & Preprocessing

Demo: 12 March 2026. Multi-dialect corpus assembled (Dhakaiya, Barishali, Sylheti, Normal, Indian Bangla); logged in `dataset_log.csv`. Preprocessed to 16 kHz mono PCM; VAD-segmented by WhisperX. Final: ≈ 4.7 h, 10 sources, $\sim 1,100$ clips.

Phase 2 — Baseline ASR Benchmark (12 models)

Demo: 2 April 2026. Whisper (S/M/L/Turbo), Wav2Vec2 variants, MMS, Vakyansh-Bn, BengaliAI Whisper/Regional, SeamlessM4T. Best result: **BengaliAI Regional** — 46.94% WER, 24.29% CER, 0.273 BLEU. Key finding: dialect speech defeats even the best open Bangla ASR models.

Phase 3 — Large Multimodal LLM Benchmark

Demo: 9 April 2026. Qwen2-Audio, Qwen3-ASR-1.7B, Voxtral-Mini, Phi-4 Multimodal, Qwen2.5-Omni (1.7B–7B range). None matched BengaliAI Regional; closest trailed by >50 percentage points WER. Conclusion: *specialisation beats scale for Bangla dialect ASR*.

Phase 4 — Fine-Tuning Methodology Study

Demo: 16 April 2026. LoRA adapter plan for Qwen3-ASR-1.7B: adapter on speech encoder; decoder weights frozen. No training run yet — this is a methodology study; training is scheduled for CSE499B.

Phase 5 — Comparative Chatbot Output Study

Selected; execution at next demo. ChatGPT, DeepSeek, Qwen, Perplexity prompted with simulated patient utterances. Outputs scored on schema consistency, field completeness, code-mixed handling. Results will inform the Module 12 report schema.

7 Conclusion

The updated 15-module design addresses every limitation of the original five-stage pipeline. The adaptive follow-up loop (Modules 6–9) ensures complete clinical profiles before assessment. Module 5 adds an emergency escalation pathway that was entirely absent before. Modules 10–11 introduce risk stratification and transparency through explainable AI. Module 15 closes the quality loop by feeding physician corrections back into model retraining. The CSE499B priority is LoRA fine-tuning of Qwen3-ASR-1.7B to reduce dialect WER to $<20\%$; once the ASR foundation is reliable, Modules 3–15 will be built and integrated iteratively.

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